THE PROBLEM

# 55 Million Potholes. $26.5B in Damage. Most Never Reported.

Reporting a civic issue takes 15–20 minutes of forms, phone calls, and wrong departments. Most citizens just give up.

**Snap it. Report it. Fix it.**

**55M**
Potholes on US roads

**$26.5B**
Annual vehicle damage

**$117B**
Annual infrastructure funding gap

**15–20**
Minutes to file a complaint today

**Issues covered:**

- Potholes
- Cracked Pavement
- Road Debris
- Broken Signs
- Faded Markings

*Sources: AAA, TRIP, National Surface Transportation Commission*

HOW CIVICLENS WORKS

# One Photo. Ten Seconds. Filed.

## 01

### Citizen Takes Photo

Open app, tap Report, snap or upload a photo of any road issue

## 02

### EfficientNetV2 Classifies

Custom model trained on 14,000+ civic images — 80% confidence threshold before proceeding

## 03

### Gemini Vision Generates

AI writes the complaint — cause, fix time, department routing. Only called if issue detected.

## 04

### Complaint Filed + Pinned

Saved to database, routed to the right city department, pinned live on the incident map

**Duplicate Detection**

Flags if same issue reported within 100m

**Live Incident Map**

Color-coded pins by severity across the city

**Department Routing**

Auto-routes to Roads, Sanitation, Electrical

**Admin Dashboard**

Real-time analytics, priority queue, status tracking

TRAINING DATASET

# Six Classes. 7,200 Images. Balanced Set.

| 7,200 | 6 | 80/10/10 |
|---|---|---|
| Total Images | Classes | Train/Val/Test |

## Potholes

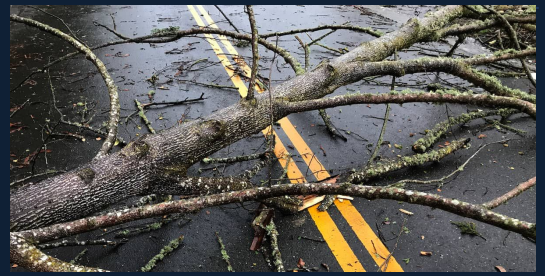
~1,200 images

## Cracked Pavement

~1,200 images

## Road Debris / Obstruction

~1,200 images



## Broken Road Signs

~1,200 images

## Faded Lane Markings

~1,200 images

## Normal Road ✓

~1,200 images

📦 Source: Roboflow Universe:
Multiple road defect datasets merged, **deduplicated** & **cleaned**

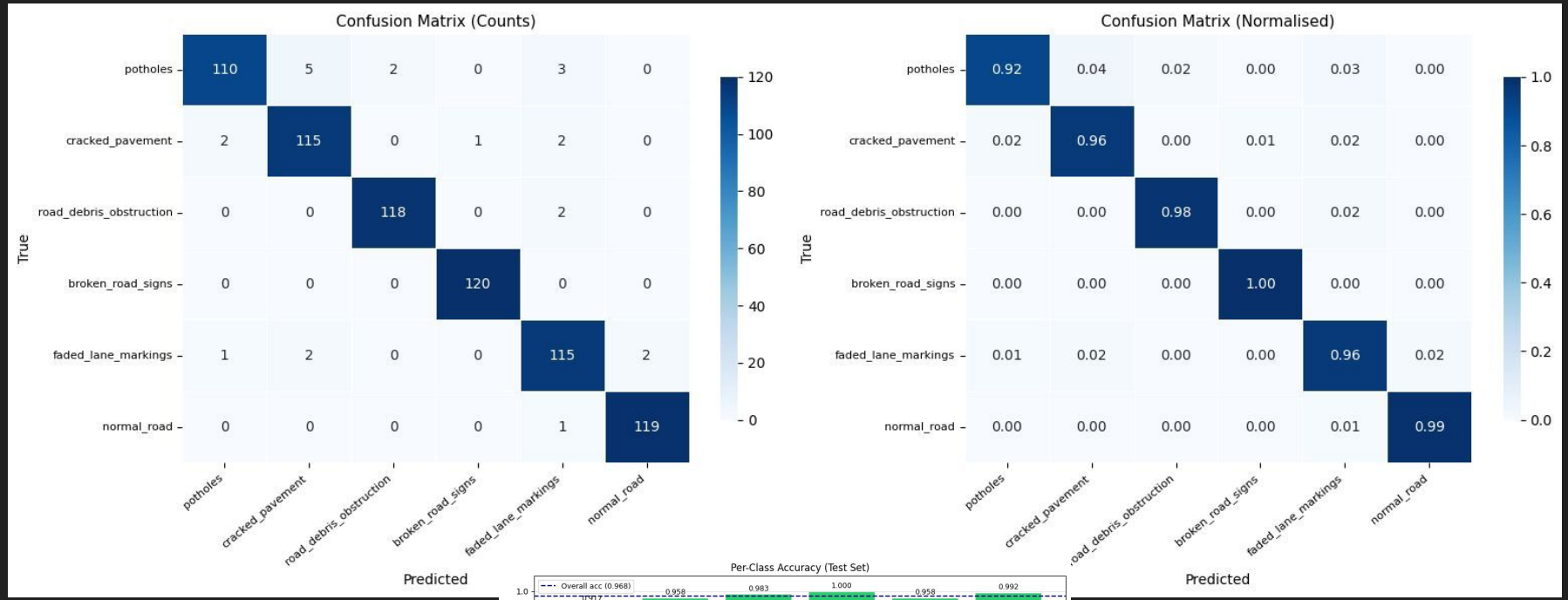
🎨 Synthetic data:
<100 bing search & manually annotated images for each class

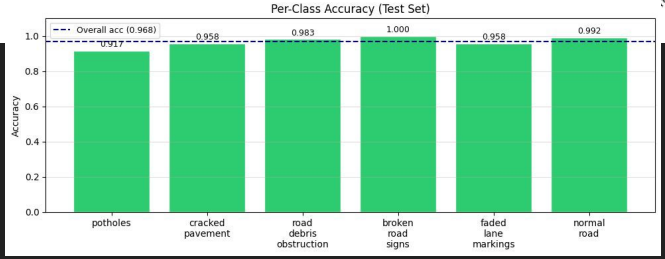
⚖️ Balanced to ~1,200 imgs/class
Corrupt & undersized files removed before training

LOSS CURVES

Training & Validation Loss
Loss progression over 8 epochs
Val spike: 2.14
2.25
1.87
1.50
1.12
0.75
0.37
0.00
Loss
Ep 1
Ep 2
Ep 3
Ep 4
Ep 5
Ep 6
Ep 7
Ep 8
Train Loss
Val Loss

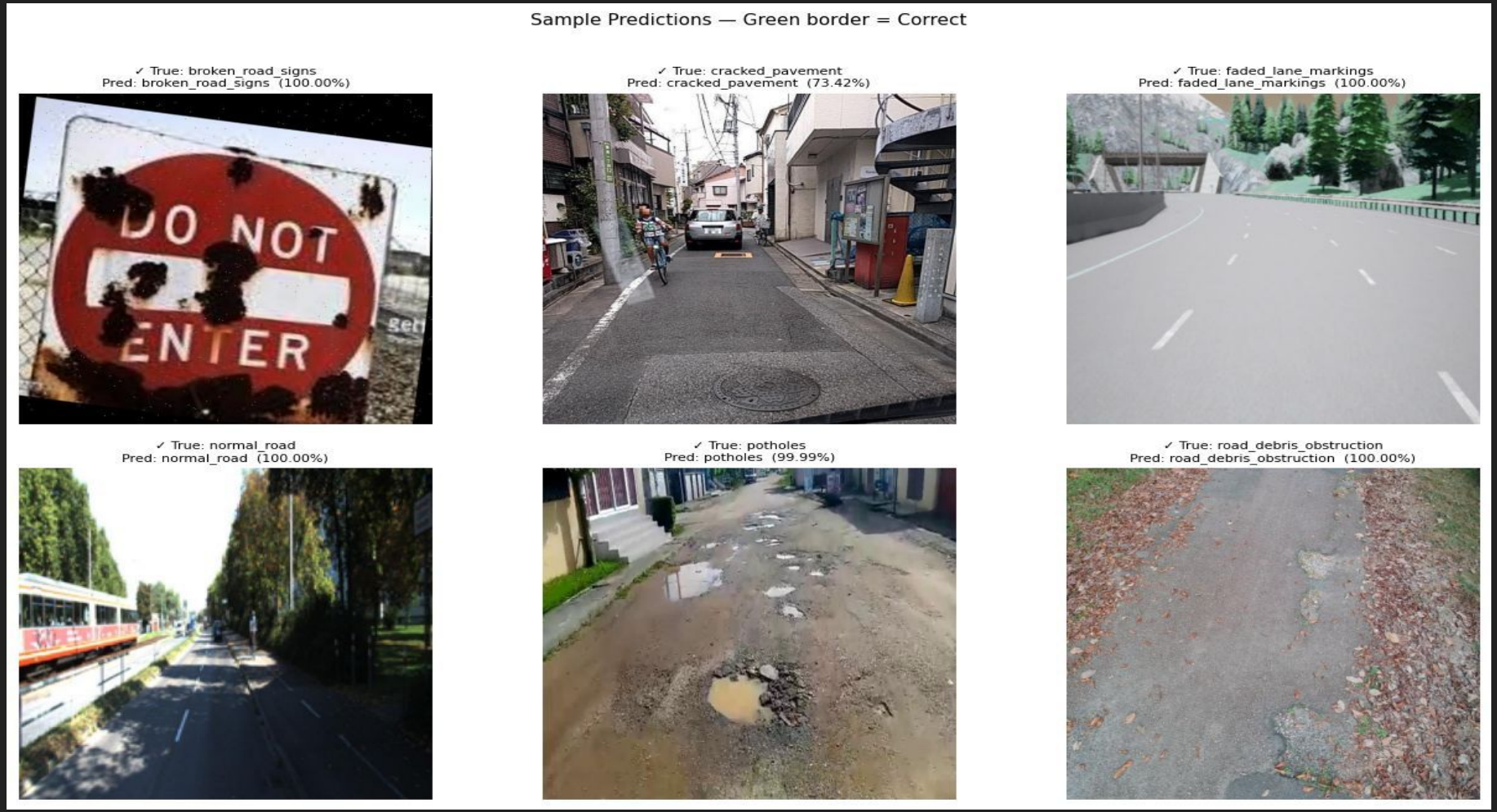
# TEST ACCURACY



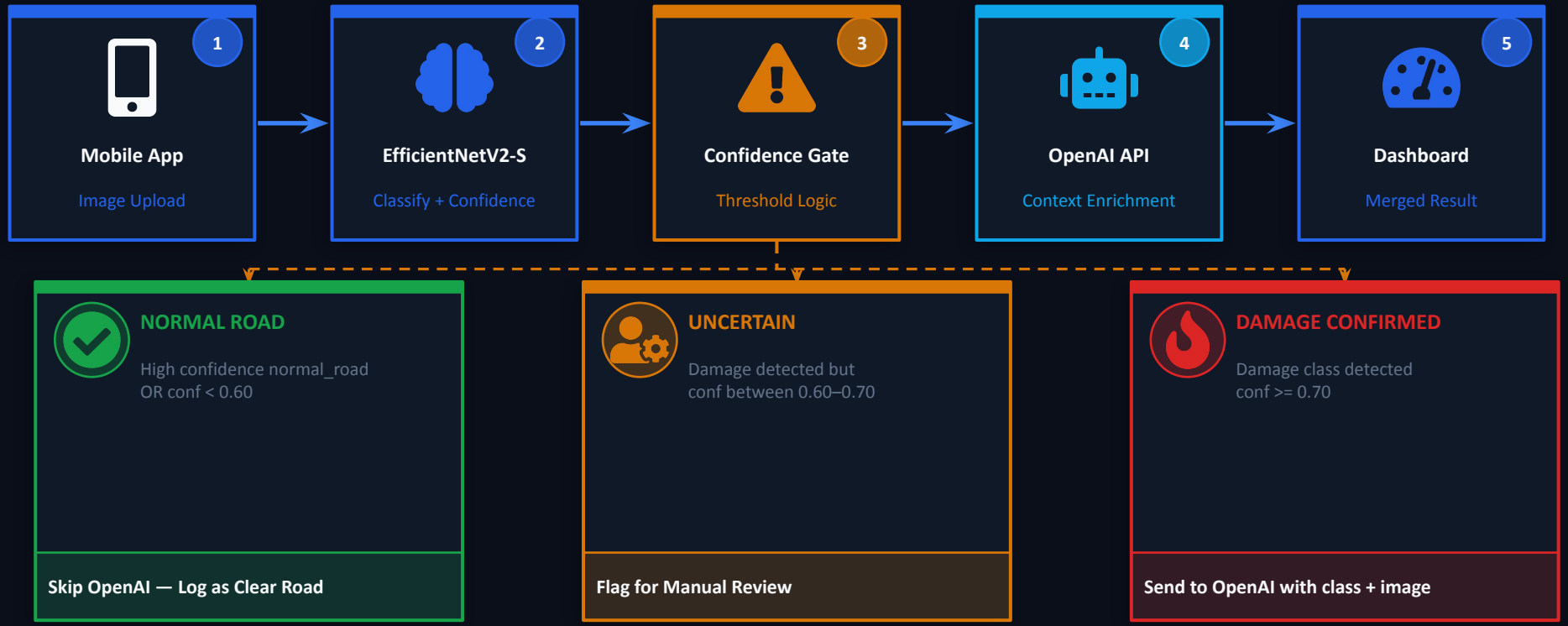




~ 800 images

## EVALUATION


Sample Predictions — Green border = Correct
✓ True: broken_road_signs
Pred: broken_road_signs (100.00%)
DO NOT
ENTER
✓ True: cracked_pavement
Pred: cracked_pavement (73.42%)
✓ True: faded_lane_markings
Pred: faded_lane_markings (100.00%)
✓ True: normal_road
Pred: normal_road (100.00%)
✓ True: potholes
Pred: potholes (99.99%)
✓ True: road_debris_obstruction
Pred: road_debris_obstruction (100.00%)

# Inference Pipeline — End-to-End Flow



# OpenAI Enrichment — Inputs & Outputs

## INPUT TO OPENAI

**Original Road Image**

Full-resolution photo from mobile app upload

**Classifier Result**

Damage class (e.g. pothole, crack, subsidence)

**Confidence Score**

Numeric score >= 0.70 from EfficientNetV2-S

*The classifier's damage type is passed directly in the prompt for grounded, context-aware analysis.*

*enriches*

## OPENAI RETURNS

**Title**

Severe pothole cluster on carriageway

**Description**

Multiple deep potholes posing structural risk

**Severity**

High — immediate action required

**Likely Cause**

Water ingress + freeze-thaw cycle

**Est. Fix Time**

4–6 hours, full patching crew

**Department**

Roads & Infrastructure Division

# Confidence Routing — Decision Thresholds

### Zone 1 — conf < 0.60

- Classified as normal road
- OpenAI call is SKIPPED
- Logged as: Clear Road
- No issue ticket created

### Zone 2 — 0.60 to 0.70

- Damage possible but uncertain
- Flagged for manual review
- Human inspector assigned
- Issue held pending verification

### Zone 3 — conf >= 0.70

- Damage class confirmed
- Image + class sent to OpenAI
- Full enrichment triggered
- Auto-filed to dashboard